Devanshu Kumar Ranjan

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Summary

Cybersecurity Engineer and AI Specialist with an M.Tech in Cyber Security from NIT Patna (CGPA: 9.06/10), specializing in SOC operations, SIEM detection engineering, network forensics, and applied machine learning for cyber defense. Architected a multi-VM Splunk SIEM enterprise lab simulating brute-force and web attack vectors with custom SPL hunting queries and real-time alerting. Analyzed packet-level DoS attack signatures in Wireshark across 6 protocols and conducted digital forensic investigations with Autopsy and FTK Imager (cryptographic hash verification and file carving). Creator of Hermes IDS (Tab2Image 2D-CNN & Random Forest with 98.6% detection accuracy on UGRansome 2024 and UNSW-NB15) and author of master's thesis on locally differentially private (LDP) federated recommendation securing user data against inference attacks. Currently engineering resilient, low-latency production diagnostic and data systems at KAS Global Commerce with Python, FastAPI, and PostgreSQL. Holder of the Google Cybersecurity Specialization.

Education

National Institute of Technology, Patna — Patna, India
M.Tech., Cyber Security (CSE) — CGPA: 9.06/10 — *August 2024 – May 2026*

Graphic Era University, Dehradun — Dehradun, India
B.Tech., Computer Science and Engineering — CGPA: 8.83/10 — *June 2019 – June 2023*

Sri Chaitanya Techno School (CBSE) — Visakhapatnam, India
Senior Secondary — 80.6% — *2019*

Cambridge School (ICSE) — Cuttack, Odisha
Secondary — 90.17% — *2017*

Experience

KAS Global Commerce Inc. — Bilaspur, HP (On-site)
AI Engineer — *July 2026 – Present*

- Architected and deployed a production LangGraph multi-agent RCA engine from scratch (research to deployment): **15 live diagnostics and 50+ root-cause rules** turning seller queries (English/Hinglish) into automated anomaly investigations; engineered input sanitization and security guardrails to prevent prompt injection; reduced PostgreSQL analytics fetch latency from $>$**1 minute down to** $<$**7 seconds (88%+ speedup)** via multi-query parallelization, compound B-tree/GIN indexing, and optimized relational joins.
- Cut monthly catalog harvesting and multimodal vision processing overhead to **zero manual intervention** via scheduled cron orchestrators and automated schema recovery.
- Scaled catalog harvesting, normalization, and vision intelligence to **9 global marketplaces** (Amazon.in, Amazon.com, ONDC, Walmart, Flipkart, eBay, etc.).
- Analyzed visual design patterns, product placement, background settings, and headline messaging across competitor Amazon A+ content to generate structured insight JSONs for automated A+ content creation.

KAS Global Commerce Inc. — Bilaspur, HP (On-site)
AI Research Intern — *September 2025 – June 2026*

- **Distributed Two-Stage Web Scraping Engine:** Engineered high-volume catalog harvesting across **Amazon.in**, **Amazon.com**, and **ONDC**; reduced full-catalog scraping time from **2–3 months down to 1.5 weeks** (90%+ latency reduction) via threaded Zyte API calls at **1,000 req/min** with MongoDB persistence.
- **Taxonomy ETL & Dense Vector Search:** Developed taxonomy breadcrumb normalization and dense semantic vector search using OpenAI `text-embedding-3-large` (3,072 dimensions) and PostgreSQL with `pgvector` (IVFFLAT indexing) for cross-marketplace category mapping, achieving **sub-50ms retrieval latency**.
- **Common API Endpoints & Chatbot RAG:** Built 3 core microservices (`category mapper`, `insights`, `product details`) enabling multi-turn conversational category RAG with dynamic chip disambiguation powering the MSC IQ assistant.
- **Multimodal Text & Image Insight Pipeline (Amazon & ONDC):** Designed foundational category intelligence combining quantitative statistical analysis with Gemini 2.5 Flash inference for Amazon and ONDC product listings.

Projects

Splunk SIEM Lab: Attack Simulation, Hunting & Security Monitoring — April 2026 – May 2026

- **Tools:** Splunk Enterprise, Splunk Universal Forwarder, Kali Linux, Ubuntu Server, Apache2, Hydra, Nikto, VMware
- Engineered a 3-VM virtualized SOC environment (Ubuntu Server, Kali Linux, Splunk Enterprise) streaming `auditd`, `auth.log`, and Apache telemetry into Splunk in real time.
- Simulated SSH brute-force attacks via Hydra (600+ failed attempts) and web vulnerability scanning with Nikto to generate attack datasets.

- Formulated custom SPL hunting queries identifying brute-force bursts, top source IPs, and 404 clusters; configured automated threshold alerts and multi-panel Security Monitoring Dashboards.

Hermes IDS: Hybrid 2D-CNN & Random Forest Intrusion Detection System May 2025 – June 2025

- **Tools:** Python, PyTorch, Scikit-learn, Tab2Image, Tkinter, Pandas
- Engineered an autonomous threat detection engine converting tabular NetFlow records into 2D grayscale feature representations (Tab2Image) for spatial feature extraction via CNN.
- Benchmarked against Random Forest and ensemble models on the **UGRansome 2024** and **UNSW-NB15** datasets, achieving **98.6% detection accuracy**.
- Developed an interactive Tkinter SOC console for real-time packet inspection, anomaly visualization, and automated threat mitigation.

DoS Attack Vector Simulation & Protocol Packet Forensics January 2025 – January 2025

- **Tools:** Kali Linux, Metasploitable 2, Wireshark, hping3, Nmap, VirtualBox
- Executed 6 distinct DoS attack vectors (Slowloris, TCP SYN Flood, UDP Flood, TCP FIN Flood, PUSH+ACK Flood, ICMP Flood) against a Metasploitable target in a bridged virtual network.
- Captured and analyzed raw PCAP traffic in Wireshark, documenting packet inter-arrival patterns, TCP handshakes, protocol-level anomaly signatures, and firewall mitigation rules.

Digital Forensic Investigation & File Recovery (Autopsy + FTK Imager) January 2025 – February 2025

- **Tools:** Autopsy, FTK Imager, Kali Linux, Raw Disk Image (.dd)
- Acquired bit-stream forensic disk images in `.dd` format using FTK Imager, verifying cryptographic hashes (MD5 and SHA-1) to ensure evidence integrity.
- Conducted incident timeline reconstruction and artifact analysis using Autopsy, inspecting file system metadata, deleted registry structures, and system event logs.
- Carved deleted files from unallocated clusters and verified file integrity using hex header and footer signature analysis (`%PDF-` to `%%EOF`).

Privacy-Preserving Federated Sequential Recommendation (M.Tech Dissertation) August 2025 – May 2026

- **Tools:** Python, PyTorch, Differential Privacy, Federated Learning, Contrastive Learning
- Formulated **LDP-SeqMask** (Locally Differentially Private Sequence Masking), injecting calibrated noise into interaction sequences locally so raw client item sequences are never shared, defending against trajectory reconstruction attacks.
- Replaced unstable Exponential Mechanism item selection with a **calibrated Laplace perturbation mechanism**, eliminating gradient divergence and stabilizing federated model updates.
- Applied **View-based Contrastive Learning (View-CL)** to sustain recommendation accuracy under heavy differential privacy noise, and integrated sequence-aware heterogeneous client model selection.

Technical Skills

SIEM & SOC Operations: Splunk Enterprise, Splunk Universal Forwarder, SPL Querying, Rule Creation, Alerting, SOC Dashboards, Incident Response, Log Analysis (`auth.log`, `syslog`, Apache)

Threat Detection & Network Security: Wireshark, Nmap, Kali Linux, Metasploit, hping3, Hydra, Nikto, Slowloris, TCP/IP Protocol Forensics, Firewall Rules (iptables), Packet Inspection

Digital Forensics & IR: Autopsy, FTK Imager, File Carving, Timeline Reconstruction, Cryptographic Hash Verification (MD5/SHA-1), Hex Analysis, Artifact Recovery

AI & Security Intersections: Locally Differential Privacy (LDP), Federated Learning, Tab2Image 2D-CNN, Adversarial Anomaly Detection, PyTorch, Scikit-learn, XGBoost

Languages & Systems: Python, Bash, C++, Linux / Ubuntu Server, VMware Workstation, VirtualBox, PostgreSQL, `pgvector`, MongoDB Atlas, Git, Docker

Certifications

Google Cybersecurity Specialization (Coursera)	December 2024 – March 2025
Certified Python Developer Associate (PDAC)	March 2024
Data Science for Engineers (NPTEL)	January 2022 – March 2022
Machine Learning (Teachnook)	February 2022 – March 2022